

Mathematical Foundations of Machine Learning

Assignment 7 Solutions

Question 1 (15 points)

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have singular values

$$\sigma_1 \geq \sigma_2 \geq \dots \geq 0,$$

and let

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

be an SVD of $\mathbf{A}$. Denote the columns of $\mathbf{V}$ by $\mathbf{v}_1, \dots, \mathbf{v}_n$.

(a) (10 points) Prove that given $\|\mathbf{x}\|_2 = 1$,

$$\|\mathbf{A}\|_2 = \max_{\mathbf{x}} \|\mathbf{A}\mathbf{x}\|_2 = \sigma_1.$$

(b) (5 points) Characterize all unit vectors $\mathbf{x}$ for which equality holds, i.e.

$$\|\mathbf{A}\mathbf{x}\|_2 = \sigma_1.$$

Solution.

(a) Let $\mathbf{x} \in \mathbb{R}^n$ satisfy $\|\mathbf{x}\|_2 = 1$. Since $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthonormal basis, we may write

$$\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i, \quad \sum_{i=1}^n c_i^2 = 1.$$

Then

$$\mathbf{V}^\top \mathbf{x} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix},$$

so

$$\mathbf{A}\mathbf{x} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{x} = \mathbf{U}\mathbf{\Sigma} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

Since $\mathbf{U}$ is orthogonal, it preserves Euclidean norm. Hence

$$\begin{aligned} \|\mathbf{A}\mathbf{x}\|_2^2 &= \left\| \mathbf{\Sigma} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \right\|_2^2 \\ &= \sum_{i=1}^n \sigma_i^2 c_i^2 \leq \sigma_1^2 \sum_{i=1}^n c_i^2 = \sigma_1^2. \end{aligned}$$

Therefore

$$\|\mathbf{A}\mathbf{x}\|_2 \leq \sigma_1 \quad \text{for all } \|\mathbf{x}\|_2 = 1,$$

so

$$\max_x \|\mathbf{A}\mathbf{x}\|_2 \leq \sigma_1.$$

On the other hand, if we choose $\mathbf{x} = \mathbf{v}_1$, then

$$\mathbf{A}\mathbf{v}_1 = \sigma_1 \mathbf{u}_1,$$

so

$$\|\mathbf{A}\mathbf{v}_1\|_2 = \sigma_1 \|\mathbf{u}_1\|_2 = \sigma_1.$$

Hence

$$\max_{\|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2 = \sigma_1.$$

Since $\|\mathbf{A}\|_2 = \max_{\|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2$, we conclude

$$\|\mathbf{A}\|_2 = \sigma_1.$$

(b) Equality

$$\|\mathbf{A}\mathbf{x}\|_2 = \sigma_1$$

holds if and only if

$$\|\mathbf{A}\mathbf{x}\|_2^2 = \sum_{i=1}^n \sigma_i^2 c_i^2 = \sigma_1^2 \sum_{i=1}^n c_i^2.$$

This happens exactly when

$$c_i = 0 \quad \text{for every } i \text{ such that } \sigma_i < \sigma_1.$$

Therefore equality holds exactly for the unit vectors in the span of the right singular vectors corresponding to the largest singular value σ_1 .

Question 2 (15 points)

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have singular values

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0,$$

with corresponding right singular vectors $\mathbf{v}_1, \dots, \mathbf{v}_r$. Fix an integer k with $1 \leq k \leq r$, and define

$$\mathbf{x}_k = \frac{1}{\sqrt{k}} \sum_{i=1}^k \mathbf{v}_i.$$

(a) (10 points) Show that $\|\mathbf{x}_k\|_2 = 1$ and compute $\|\mathbf{A}\mathbf{x}_k\|_2^2$. (Hint: you may need to use the singular value equation. $\|\mathbf{A}\mathbf{x}_k\|_2^2$ means the square of the 2-norm of the vector $\mathbf{A}\mathbf{x}_k$)

(b) (5 points) Deduce that

$$\|\mathbf{A}\|_2 \geq \sqrt{\frac{1}{k} \sum_{i=1}^k \sigma_i^2}.$$

Solution.

(a) Since $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is an orthonormal set,

$$\|\mathbf{x}_k\|_2^2 = \left\| \frac{1}{\sqrt{k}} \sum_{i=1}^k \mathbf{v}_i \right\|_2^2 = \frac{1}{k} \sum_{i=1}^k \|\mathbf{v}_i\|_2^2 = \frac{1}{k} \cdot k = 1.$$

Hence $\|\mathbf{x}_k\|_2 = 1$.

Next,

$$\mathbf{A}\mathbf{x}_k = \frac{1}{\sqrt{k}} \sum_{i=1}^k \mathbf{A}\mathbf{v}_i = \frac{1}{\sqrt{k}} \sum_{i=1}^k \sigma_i \mathbf{u}_i,$$

where $\mathbf{u}_i$ are the corresponding left singular vectors. Since $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is orthonormal,

$$\|\mathbf{A}\mathbf{x}_k\|_2^2 = \left\| \frac{1}{\sqrt{k}} \sum_{i=1}^k \sigma_i \mathbf{u}_i \right\|_2^2 = \frac{1}{k} \sum_{i=1}^k \sigma_i^2.$$

(b) Since $\|\mathbf{x}_k\|_2 = 1$, by the definition of spectral norm,

$$\|\mathbf{A}\|_2 = \max_{\|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2 \geq \|\mathbf{A}\mathbf{x}_k\|_2.$$

Using part (a),

$$\|\mathbf{A}\mathbf{x}_k\|_2 = \sqrt{\frac{1}{k} \sum_{i=1}^k \sigma_i^2}.$$

Therefore

$$\|\mathbf{A}\|_2 \geq \sqrt{\frac{1}{k} \sum_{i=1}^k \sigma_i^2}.$$

Question 3 (20 points)

Three players — Alice, Bob, and Charlie — compete in a tournament. Their scores in the first two games are recorded in the matrix

$$\mathbf{R} = \begin{bmatrix} 12 & 4 \\ 0 & 12 \\ 4 & 0 \end{bmatrix},$$

where rows correspond to Alice, Bob, and Charlie, respectively, and columns correspond to Game 1 and Game 2.

(a) (12 points) Compute an SVD

$$\mathbf{R} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

with $\mathbf{U} \in \mathbb{R}^{3 \times 3}$ orthogonal, $\mathbf{V} \in \mathbb{R}^{2 \times 2}$ orthogonal, and $\mathbf{\Sigma} \in \mathbb{R}^{3 \times 2}$ diagonal.

(b) (4 points) Using your SVD, find the best (in terms of spectral norm) rank-1 approximation.

(c) (4 points) Based on your rank-1 approximation, what insight can you draw about the dominant scoring pattern of the three players across the first two games?

Solution.

(a) First compute

$$\mathbf{R}^\top \mathbf{R} = \begin{bmatrix} 12 & 0 & 4 \\ 4 & 12 & 0 \end{bmatrix} \begin{bmatrix} 12 & 4 \\ 0 & 12 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} 160 & 48 \\ 48 & 160 \end{bmatrix} = 16 \begin{bmatrix} 10 & 3 \\ 3 & 10 \end{bmatrix}.$$

The eigenvalues are

$$\lambda_1 = 208, \quad \lambda_2 = 112,$$

with corresponding orthonormal eigenvectors

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Hence the singular values are

$$\sigma_1 = 4\sqrt{13}, \quad \sigma_2 = 4\sqrt{7}.$$

Let

$$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Now compute the left singular vectors:

$$\mathbf{u}_1 = \frac{1}{\sigma_1} \mathbf{R} \mathbf{v}_1 = \frac{1}{4\sqrt{13}} \mathbf{R} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{26}} \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix},$$

and

$$\mathbf{u}_2 = \frac{1}{\sigma_2} \mathbf{R} \mathbf{v}_2 = \frac{1}{4\sqrt{7}} \mathbf{R} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{14}} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}.$$

Complete these to an orthonormal basis of $\mathbb{R}^3$ with

$$\mathbf{u}_3 = \frac{1}{\sqrt{91}} \begin{bmatrix} 3 \\ -1 \\ -9 \end{bmatrix}.$$

Thus

$$\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3] = \begin{bmatrix} \frac{4}{\sqrt{26}} & \frac{2}{\sqrt{14}} & \frac{3}{\sqrt{91}} \\ \frac{3}{\sqrt{26}} & -\frac{3}{\sqrt{14}} & -\frac{1}{\sqrt{91}} \\ \frac{1}{\sqrt{26}} & \frac{1}{\sqrt{14}} & -\frac{9}{\sqrt{91}} \end{bmatrix}, \quad \mathbf{\Sigma} = \begin{bmatrix} 4\sqrt{13} & 0 \\ 0 & 4\sqrt{7} \\ 0 & 0 \end{bmatrix}.$$

Therefore

$$\mathbf{R} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top.$$

(b) The rank-1 approximation obtained by keeping only the largest singular value is

$$\hat{\mathbf{R}}_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top = 4\sqrt{13} \left(\frac{1}{\sqrt{26}} \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix} \right) \left(\frac{1}{\sqrt{2}} [1 \quad 1] \right).$$

Since

$$\frac{4\sqrt{13}}{\sqrt{26}\sqrt{2}} = 2,$$

we get

$$\hat{\mathbf{R}}_1 = 2 \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix} [1 \quad 1] = \begin{bmatrix} 8 & 8 \\ 6 & 6 \\ 2 & 2 \end{bmatrix}.$$

(c) The rank-1 approximation says that the dominant pattern is explained by a single overall scoring-strength factor. In this approximation, the two game columns are identical, so the main effect is not the difference between Game 1 and Game 2, but an overall ranking of the players.

The dominant ordering is

$$\text{Alice} > \text{Bob} > \text{Charlie}.$$

So most of the variation in the original score table is captured by one common performance level, with Alice clearly strongest overall, Bob next, and Charlie last.

Question 4 (15 points)

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be invertible, with singular values

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0.$$

(a) (9 points) Prove that for every $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{Ax}\|_2 \geq \sigma_n \|\mathbf{x}\|_2.$$

(b) (6 points) Deduce that

$$\|\mathbf{A}^{-1}\|_2 = \frac{1}{\sigma_n}.$$

Solution.

(a) Let

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

be an SVD of $\mathbf{A}$. Since $\mathbf{V}$ is orthogonal, for any $\mathbf{x} \in \mathbb{R}^n$ we may write

$$\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i.$$

Then

$$\|\mathbf{x}\|_2^2 = \sum_{i=1}^n c_i^2,$$

and

$$\mathbf{Ax} = \sum_{i=1}^n c_i \sigma_i \mathbf{u}_i.$$

Therefore

$$\|\mathbf{Ax}\|_2^2 = \sum_{i=1}^n \sigma_i^2 c_i^2 \geq \sigma_n^2 \sum_{i=1}^n c_i^2 = \sigma_n^2 \|\mathbf{x}\|_2^2.$$

Taking square roots gives

$$\|\mathbf{Ax}\|_2 \geq \sigma_n \|\mathbf{x}\|_2.$$

(b) Let $\mathbf{y} \in \mathbb{R}^n$. Since $\mathbf{A}$ is invertible, we can write $\mathbf{y} = \mathbf{Ax}$ where $\mathbf{x} = \mathbf{A}^{-1}\mathbf{y}$. Applying part (a) to $\mathbf{x}$ gives

$$\|\mathbf{y}\|_2 = \|\mathbf{Ax}\|_2 \geq \sigma_n \|\mathbf{x}\|_2 = \sigma_n \|\mathbf{A}^{-1}\mathbf{y}\|_2.$$

Hence

$$\|\mathbf{A}^{-1}\mathbf{y}\|_2 \leq \frac{1}{\sigma_n} \|\mathbf{y}\|_2.$$

Taking the maximum over all nonzero $\mathbf{y}$ yields

$$\|\mathbf{A}^{-1}\|_2 \leq \frac{1}{\sigma_n}.$$

To show equality, take $\mathbf{y} = \mathbf{u}_n$. Since

$$\mathbf{Av}_n = \sigma_n \mathbf{u}_n,$$

we have

$$\mathbf{A}^{-1}\mathbf{u}_n = \frac{1}{\sigma_n} \mathbf{v}_n.$$

Therefore

$$\|\mathbf{A}^{-1}\mathbf{u}_n\|_2 = \frac{1}{\sigma_n}.$$

Since $\|\mathbf{u}_n\|_2 = 1$, this implies

$$\|\mathbf{A}^{-1}\|_2 \geq \frac{1}{\sigma_n}.$$

Combining the two inequalities gives

$$\|\mathbf{A}^{-1}\|_2 = \frac{1}{\sigma_n}.$$

Question 5 (15 points)

Consider the transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$u = x^2 + yz, \quad v = e^y + xz, \quad w = xy + z^2.$$

(a) (9 points) Compute the Jacobian matrix

$$\mathbf{J}(x, y, z).$$

(b) (6 points) Compute $\det(\mathbf{J}(x, y, z))$.

Solution.

(a) The Jacobian matrix is

$$\mathbf{J}(x, y, z) = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix}.$$

Compute each derivative:

$$\begin{aligned} \frac{\partial u}{\partial x} &= 2x, & \frac{\partial u}{\partial y} &= z, & \frac{\partial u}{\partial z} &= y, \\ \frac{\partial v}{\partial x} &= z, & \frac{\partial v}{\partial y} &= e^y, & \frac{\partial v}{\partial z} &= x, \\ \frac{\partial w}{\partial x} &= y, & \frac{\partial w}{\partial y} &= x, & \frac{\partial w}{\partial z} &= 2z. \end{aligned}$$

Therefore

$$\mathbf{J}(x, y, z) = \begin{bmatrix} 2x & z & y \\ z & e^y & x \\ y & x & 2z \end{bmatrix}.$$

(b) Expanding along the first row,

$$\begin{aligned} \det(\mathbf{J}) &= 2x \begin{vmatrix} e^y & x \\ x & 2z \end{vmatrix} - z \begin{vmatrix} z & x \\ y & 2z \end{vmatrix} + y \begin{vmatrix} z & e^y \\ y & x \end{vmatrix} \\ &= 2x(2ze^y - x^2) - z(2z^2 - xy) + y(xz - ye^y). \end{aligned}$$

Simplifying,

$$\det(\mathbf{J}(x, y, z)) = 4xze^y - 2x^3 - 2z^3 + 2xyz - y^2e^y.$$

Question 6 (10 points)

Let

$$\mathbf{M} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \mathbf{z} = \mathbf{M}\mathbf{x} + \mathbf{b}, \quad \mathbf{x} \in \mathbb{R}^2.$$

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$f(\mathbf{x}) = \begin{bmatrix} z_1^2 + z_2 \\ z_1 e^{z_2} \end{bmatrix}, \quad \text{where } \mathbf{z} = \mathbf{M}\mathbf{x} + \mathbf{b}.$$

- (a) (6 points) Compute $\frac{\partial f}{\partial \mathbf{x}}$.
- (b) (4 points) Evaluate $\frac{\partial f}{\partial \mathbf{x}}$ at

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Solution.

- (a) We first compute

$$\frac{\partial f}{\partial \mathbf{z}} = \begin{bmatrix} 2z_1 & 1 \\ e^{z_2} & z_1 e^{z_2} \end{bmatrix}, \quad \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \mathbf{M} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Hence

$$\frac{\partial f}{\partial \mathbf{x}} = \frac{\partial f}{\partial \mathbf{z}} \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \begin{bmatrix} 2z_1 & 1 \\ e^{z_2} & z_1 e^{z_2} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2z_1 & 4z_1 + 1 \\ e^{z_2} & (z_1 + 2)e^{z_2} \end{bmatrix}.$$

- (b) At

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

we have

$$\mathbf{z} = \mathbf{M}\mathbf{x} + \mathbf{b} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Therefore

$$\left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}=\begin{bmatrix} 1 & 0 \end{bmatrix}^\top} = \begin{bmatrix} 4 & 9 \\ e^{-1} & 4e^{-1} \end{bmatrix}.$$

Question 7 (10 points)

Let

$$g(\mathbf{s}) = (\mathbf{x} - \mathbf{A}\mathbf{s})^\top \mathbf{W}(\mathbf{x} - \mathbf{A}\mathbf{s}),$$

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{n \times d}$, $\mathbf{s} \in \mathbb{R}^d$, and $\mathbf{W} \in \mathbb{R}^{n \times n}$ is symmetric.

- (a) (3 points) Expand $g(\mathbf{s})$ into constant, linear, and quadratic terms in $\mathbf{s}$.
- (b) (4 points) Derive the gradient $\frac{\partial g}{\partial \mathbf{s}}$.
- (c) (3 points) A stationary point of g is a vector $\mathbf{s}$ such that

$$\frac{\partial g}{\partial \mathbf{s}} = \mathbf{0}^\top.$$

Let

$$\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Find the stationary point $\mathbf{s}$.

Solution.

- (a) Expanding,

$$\begin{aligned} g(\mathbf{s}) &= (\mathbf{x} - \mathbf{A}\mathbf{s})^\top \mathbf{W}(\mathbf{x} - \mathbf{A}\mathbf{s}) \\ &= \mathbf{x}^\top \mathbf{W} \mathbf{x} - \mathbf{x}^\top \mathbf{W} \mathbf{A} \mathbf{s} - \mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{x} + \mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A} \mathbf{s}. \end{aligned}$$

Since $\mathbf{W}$ is symmetric, the two middle terms are equal scalars. Hence

$$g(\mathbf{s}) = \mathbf{x}^\top \mathbf{W} \mathbf{x} - 2\mathbf{x}^\top \mathbf{W} \mathbf{A} \mathbf{s} + \mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A} \mathbf{s}.$$

(b) Differentiate term by term:

$$\frac{\partial}{\partial \mathbf{s}}(\mathbf{x}^\top \mathbf{W} \mathbf{x}) = \mathbf{0}^\top,$$

$$\frac{\partial}{\partial \mathbf{s}}(-2\mathbf{x}^\top \mathbf{W} \mathbf{A} \mathbf{s}) = -2\mathbf{x}^\top \mathbf{W} \mathbf{A},$$

and

$$\frac{\partial}{\partial \mathbf{s}}(\mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A} \mathbf{s}) = \mathbf{s}^\top (\mathbf{A}^\top \mathbf{W} \mathbf{A} + (\mathbf{A}^\top \mathbf{W} \mathbf{A})^\top).$$

Since $\mathbf{W}$ is symmetric,

$$(\mathbf{A}^\top \mathbf{W} \mathbf{A})^\top = \mathbf{A}^\top \mathbf{W} \mathbf{A},$$

so

$$\frac{\partial}{\partial \mathbf{s}}(\mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A} \mathbf{s}) = 2\mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A}.$$

Therefore

$$\frac{\partial g}{\partial \mathbf{s}} = -2\mathbf{x}^\top \mathbf{W} \mathbf{A} + 2\mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A}.$$

Equivalently,

$$\frac{\partial g}{\partial \mathbf{s}} = -2(\mathbf{x} - \mathbf{A} \mathbf{s})^\top \mathbf{W} \mathbf{A}.$$

(c) A stationary point satisfies

$$\frac{\partial g}{\partial \mathbf{s}} = \mathbf{0}^\top.$$

Using part (b), this gives

$$-2\mathbf{x}^\top \mathbf{W} \mathbf{A} + 2\mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A} = \mathbf{0}^\top.$$

Hence

$$\mathbf{s}^\top \mathbf{A}^\top \mathbf{W} \mathbf{A} = \mathbf{x}^\top \mathbf{W} \mathbf{A}.$$

Taking transpose,

$$\mathbf{A}^\top \mathbf{W} \mathbf{A} \mathbf{s} = \mathbf{A}^\top \mathbf{W} \mathbf{x}.$$

Now compute

$$\mathbf{W} \mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix},$$

so

$$\mathbf{A}^\top \mathbf{W} \mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}.$$

Also,

$$\mathbf{W} \mathbf{x} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad \mathbf{A}^\top \mathbf{W} \mathbf{x} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Therefore

$$\begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} \mathbf{s} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Writing $\mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}$, we get

$$3s_1 + s_2 = 5, \quad s_1 + s_2 = 1.$$

Subtracting gives $2s_1 = 4$, so $s_1 = 2$, and then $s_2 = -1$. Hence

$$\mathbf{s} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Bonus (5 points)

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have singular values

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

Let

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$$

be an SVD of $\mathbf{A}$. Fix an integer k with $1 \leq k < r$.

Prove that for every matrix $\mathbf{B}$ with $\text{rank}(\mathbf{B}) \leq k$,

$$\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1}.$$

Solution. Let

$$\mathcal{V} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k+1}\}.$$

Then $\dim(\mathcal{V}) = k + 1$.

Since $\text{rank}(\mathbf{B}) \leq k$, the restriction of $\mathbf{B}$ to $\mathcal{V}$ cannot be one-to-one. Therefore there exists a nonzero vector $\mathbf{x} \in \mathcal{V}$ such that

$$\mathbf{B}\mathbf{x} = \mathbf{0}.$$

Normalize it so that $\|\mathbf{x}\|_2 = 1$. Since $\mathbf{x} \in \mathcal{V}$, we can write

$$\mathbf{x} = \sum_{i=1}^{k+1} c_i \mathbf{v}_i, \quad \sum_{i=1}^{k+1} c_i^2 = 1.$$

Now

$$\|\mathbf{A} - \mathbf{B}\|_2 \geq \|(\mathbf{A} - \mathbf{B})\mathbf{x}\|_2 = \|\mathbf{A}\mathbf{x}\|_2,$$

because $\mathbf{B}\mathbf{x} = \mathbf{0}$.

Also,

$$\mathbf{A}\mathbf{x} = \sum_{i=1}^{k+1} c_i \sigma_i \mathbf{u}_i,$$

so

$$\|\mathbf{A}\mathbf{x}\|_2^2 = \sum_{i=1}^{k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2 \sum_{i=1}^{k+1} c_i^2 = \sigma_{k+1}^2.$$

Hence

$$\|\mathbf{A}\mathbf{x}\|_2 \geq \sigma_{k+1}.$$

Therefore

$$\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1}.$$

This proves the claim.